

Midterm 2 Practice Problems

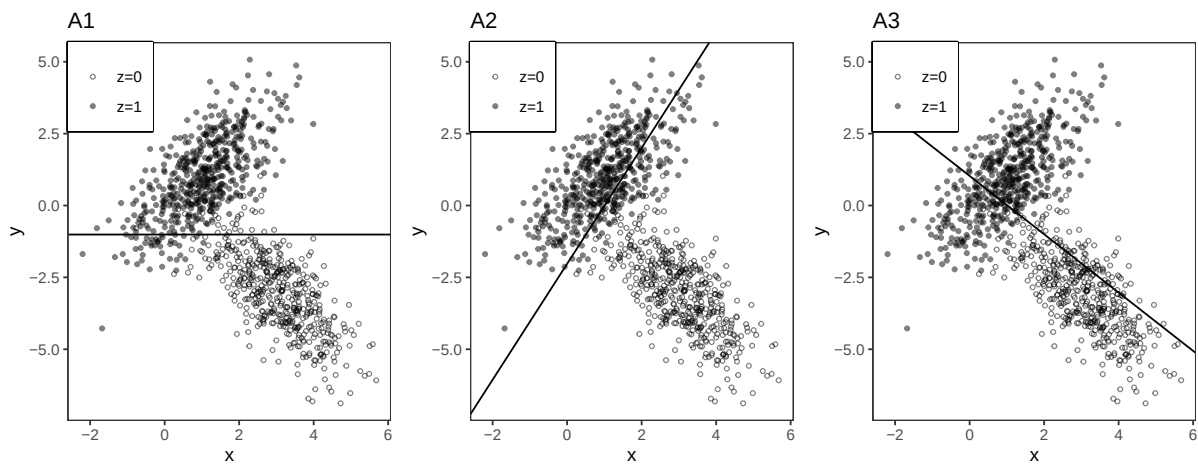
Problem 1

In the following three parts, choose the figure with the appropriate regression line(s).

Part A

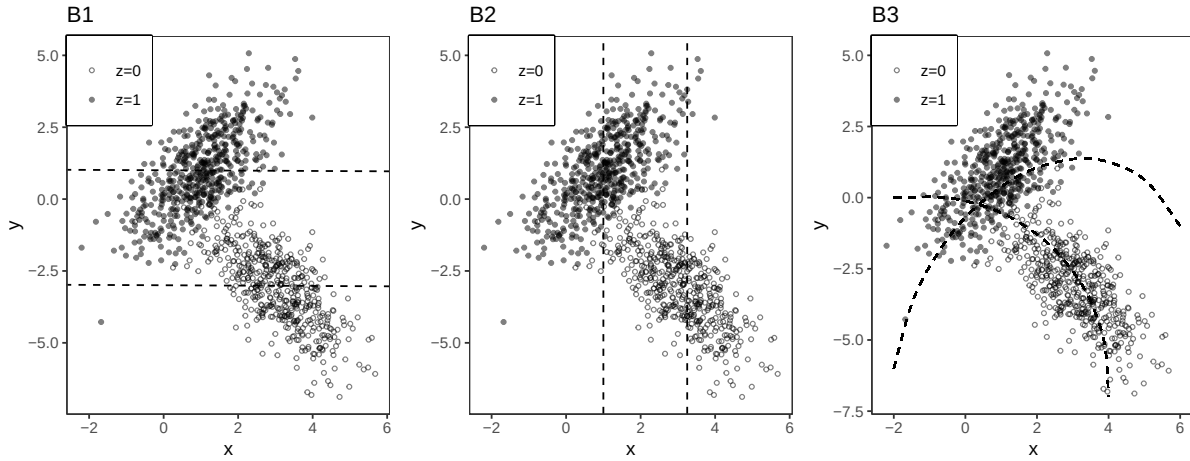
Choose the figure with the solid line that represents the simple regression line ($y \sim x$).

Warning: The ``size`` argument of ``element_rect()`` is deprecated as of ggplot2 3.4.0.
i Please use the ``linewidth`` argument instead.



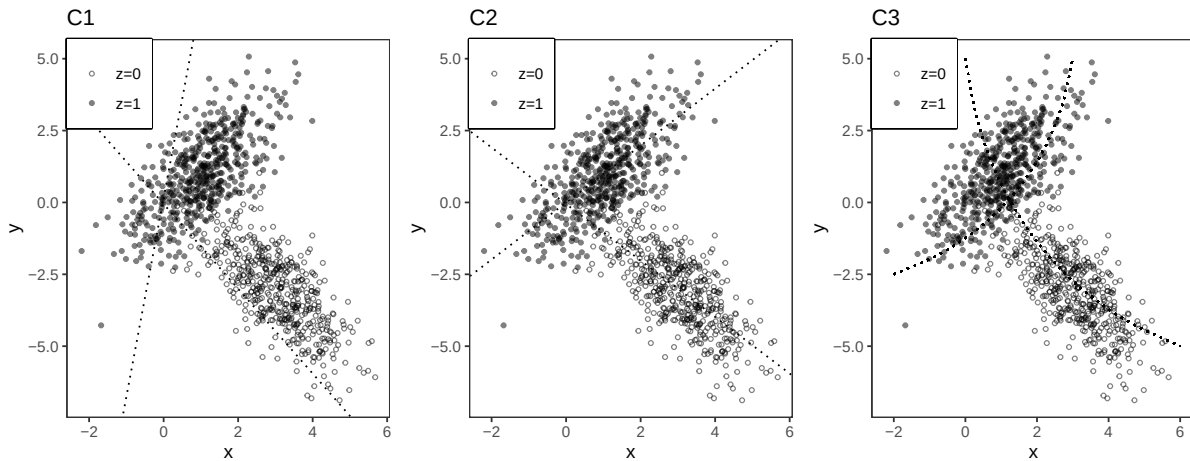
Part B

Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



Part C

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



Problem 2

Suppose we have a sample $Y_1 \dots Y_n$ stored on an **R** vector Y of length n . We run this code.

```

B=10000
Y.bar = rep(NA,B)
for(b in 1:B) {
  Y.bar[b]= mean(sample(Y,size=n,replace=TRUE))
}

```

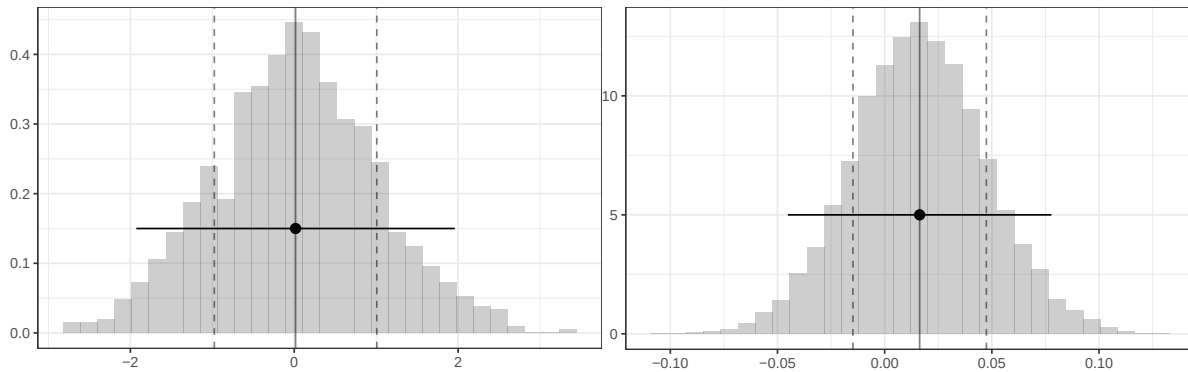


Figure 1: Histogram of the vector Y's elements Figure 2: Histogram of the Y.bar's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```

mean(Y)

```

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```

sd(Y)

```

Part c

What does this code calculate? State it in words and mark it with an ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y.bar)
```

Part d

What does this code calculate? State it in words and mark it with an ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y) - 1.96* sd(Y)/sqrt(n)
mean(Y) + 1.96* sd(Y)/sqrt(n)
```

Part e

How could you calculate roughly the same thing as in part d using ‘sd(Y.bar)’ and not ‘sd(Y)’?

Problem 3

Consider the following R code:

```
1. lm(residuals(lm(y~z)) ~ residuals(lm(z~x)))
2. lm(y ~ residuals(lm(z~x)))
3. lm(residuals(lm(y~z)) ~ residuals(lm(x~z)))
4. lm(residuals(lm(y~z)) ~ x)
5. lm(residuals(lm(y~x)) ~ residuals(lm(x~z)))
6. lm(residuals(lm(y~x)) ~ z)
7. lm(residuals(lm(y~x)) ~ residuals(lm(z~x)))
8. lm(y ~ residuals(lm(x~z)))
```

- a) Which one or more of the above will produce the coefficient on x from the multiple regression of $y \sim x + z$?
- b) Which one or more of the above will produce the same residuals as the residuals from the multiple regression of $y \sim x + z$?

Problem 4

Y_i is a individual i 's income and X_i is that individual's number of years of education. We take a random sample of individuals and record their years of education and income. Define $\hat{\mu}(16)$ as the subsample mean income among those with 16 years of education. Define an indicator function for $x = 16$ and show that $\hat{\mu}(16)$ is an unbiased estimator of $\mu(16)$, the subpopulation mean income for those with 16 years of education. Justify your steps.